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GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Retail Sales & Profit Performance Dashboard

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Experiment No.: 3

Retail Sales & Profit Performance Dashboard

GitHub-Repo-Link: [prakruthidevanga/Business-Analytics: Business Analytics](https://github.com/prakruthidevanga/Business-Analytics: Business Analytics)

1. Problem Statement

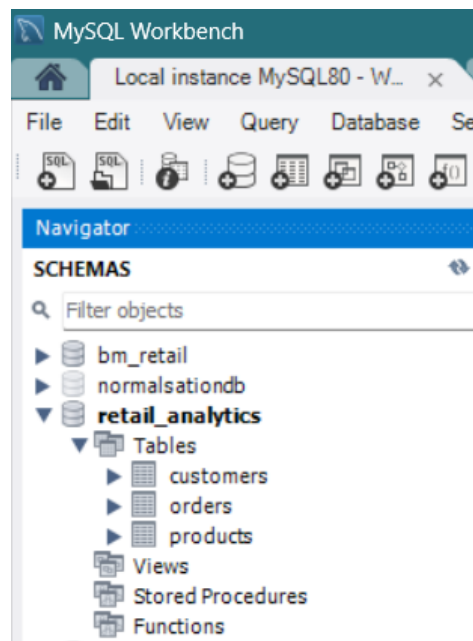
A retail business stores its transactional information in different sources, including a MySQL database and an external Kaggle CSV dataset. Analysing these sources separately makes it difficult to obtain a centralized view of sales performance, customer behaviour, product performance, and regional trends.

The objective of this experiment is to create a MySQL retail database, populate it with customer, product, and order information, import a Kaggle flat-file dataset into Power BI, connect both data sources, perform data cleaning and transformation, establish relationships between the tables, and develop an interactive Power BI dashboard for business analysis.

2. Methodology / Steps Followed

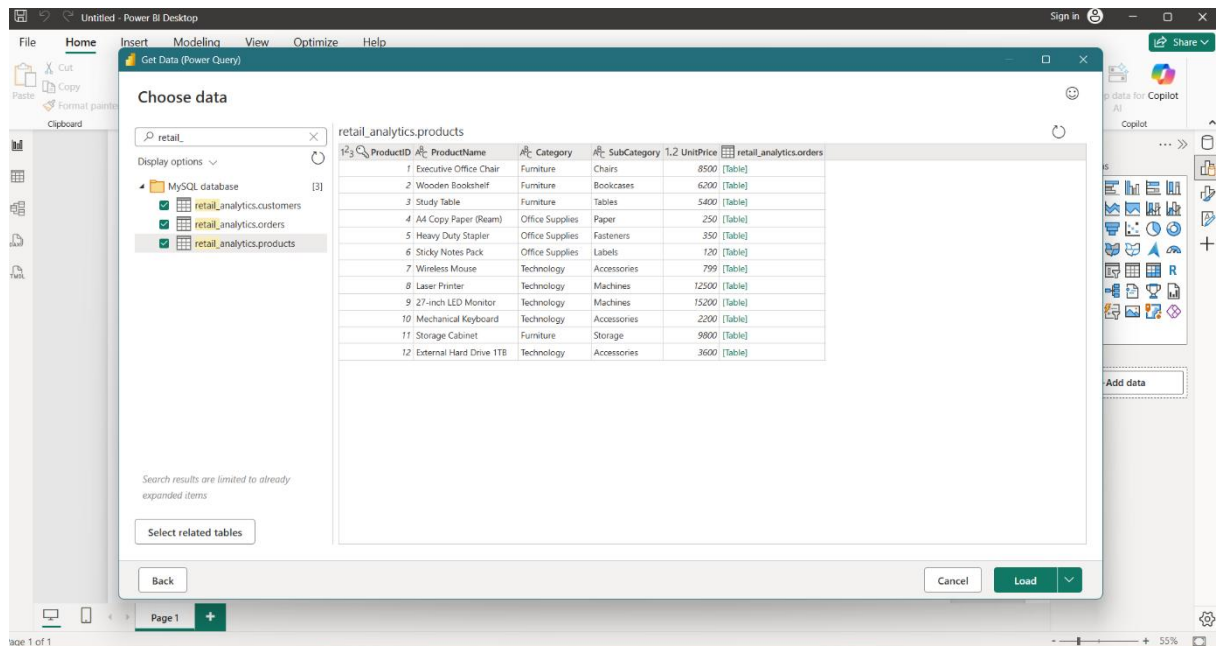
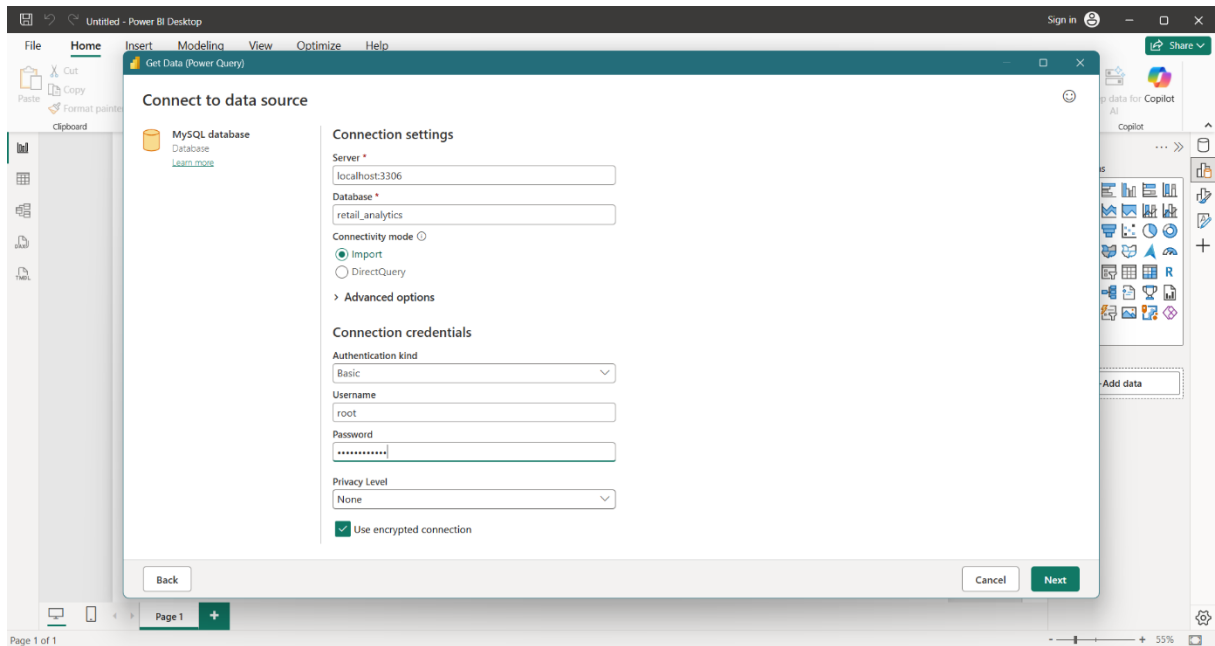
Step 1 and 2: Data Collection and MySQL Database Setup

- Data sources: MySQL database "retail_analytics" (Customers, Products, Orders tables) + Kaggle "Sample Superstore" CSV.
- Installed MySQL Server/Workbench and connected to the local server (root user).
- Ran CREATE DATABASE/CREATE TABLE and INSERT scripts to build and populate the schema.
- Verified data using SELECT * FROM Customers / Products / Orders.



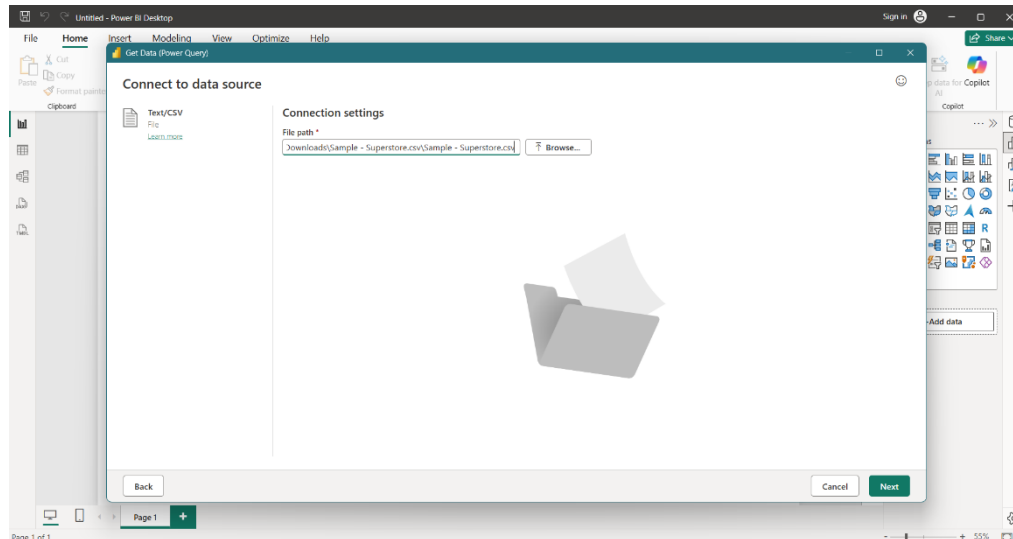
Step 3: Connecting MySQL to Power BI

- Installed the MySQL Connector/NET required by Power BI.
- Power BI Desktop → Get Data → More → MySQL database → Connect.
- Entered server (localhost:3306) and database name (retail_analytics), chose Import mode.
- Selected Customers, Products, and Orders tables in the Navigator and loaded them.



Step 4: Importing the Kaggle CSV

- Downloaded the Superstore dataset from Kaggle as a .csv file.
- Power BI Desktop → Get Data → Text/CSV → selected the file.
- Reviewed the auto-detected delimiter, columns, and data types in the preview.
- Loaded the file; it appeared as a new table alongside the MySQL tables.



Step 5: Data Cleaning & Transformation (Power Query)

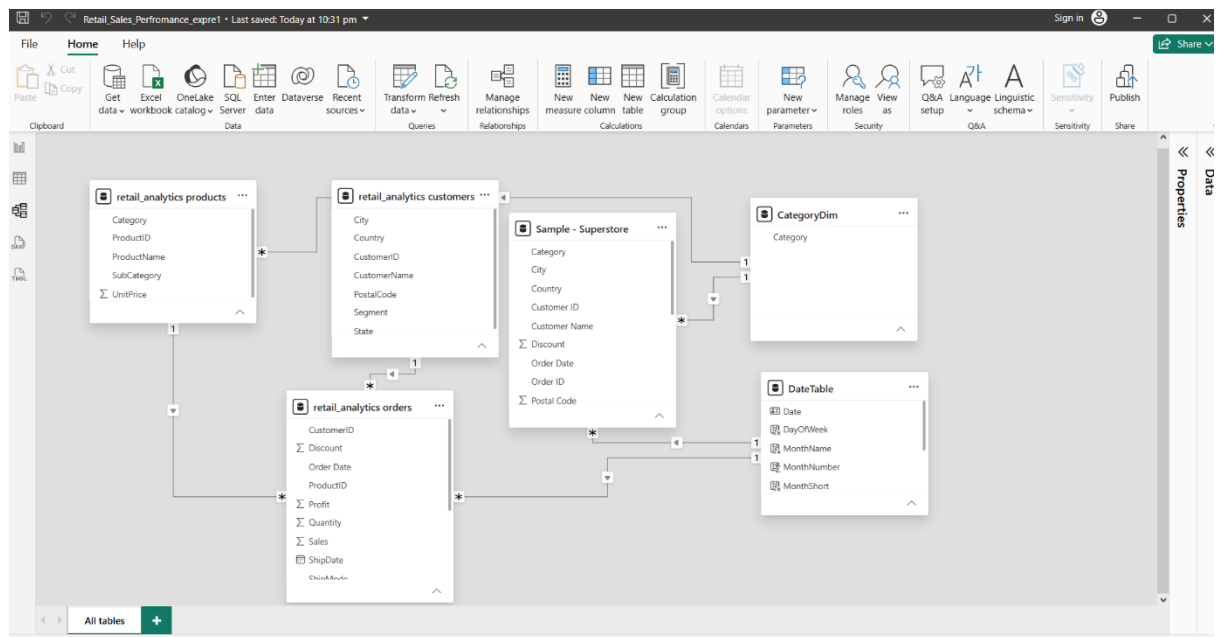
- Removed duplicate rows and handled missing/null values.
- Renamed columns to business-friendly names and corrected data types.
- Split/merged columns where needed (e.g., City/State).

The screenshot shows the Power Query Editor with a query named 'Table.Distinct(#\"Changed column type\", {\"Row ID\"})'. The table contains 21 rows of data. The columns are: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The 'Applied Steps' pane on the right shows 'Removed Duplicates'.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
12	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman
13	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480	Andrew Allen
14	CA-2016-161389	05-12-2016	10-12-2016	Standard Class	IM-15070	Irene Maddox
15	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlian
16	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlian
17	CA-2014-105893	11-11-2014	18-11-2014	Standard Class	PK-15075	Pete Kriz
18	CA-2014-167164	13-05-2014	15-05-2014	Second Class	AG-10270	Alejandro Grove
19	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli
20	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli
21	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli

Step 6: Data Modeling

- Created one-to-many relationships: Customers→Orders and Products→Orders.
- Built a dedicated DateTable and marked it as the official Date table for time intelligence.
- Linked DateTable to both Orders and the Superstore file so one date slicer filters both sources.
- Created a shared CategoryDim table to connect the flat file to the model via Category, forming a star schema.



Step 7: DAX Measures

- Total Sales = Total Sales = SUM ('retail_analytics orders'[Sales])
- Total Profit = Total Profit = SUM ('retail_analytics orders'[Profit])
- Total Orders = Total Orders = COUNTROWS ('retail_analytics orders')
- Profit Margin % = DIVIDE (Total Profit, Total Sales).
- Avg Order Value = DIVIDE (Total Sales, Total Orders).
- YoY Sales Growth % using SAMEPERIODLASTYEAR (currently blank — only Jan–Jun 2024 data available) = VAR CurrentSales = [Total Sales] VAR PriorYearSales = CALCULATE ([Total Sales], SAMEPERIODLASTYEAR ('DateTable'[Date])) RETURN DIVIDE (CurrentSales - PriorYearSales, PriorYearSales, 0)

Step 8: Dashboard Development

- KPI cards for Total Sales, Total Profit, Total Orders, and Profit Margin %.
- Customer Analysis: donut chart by Segment + customers bar chart.
- Product Analysis: category/sub-category chart + products bar chart.
- State map, monthly trend line chart, and slicers for Year/Month, Category, State, and Ship Mode.
- Applied a consistent theme

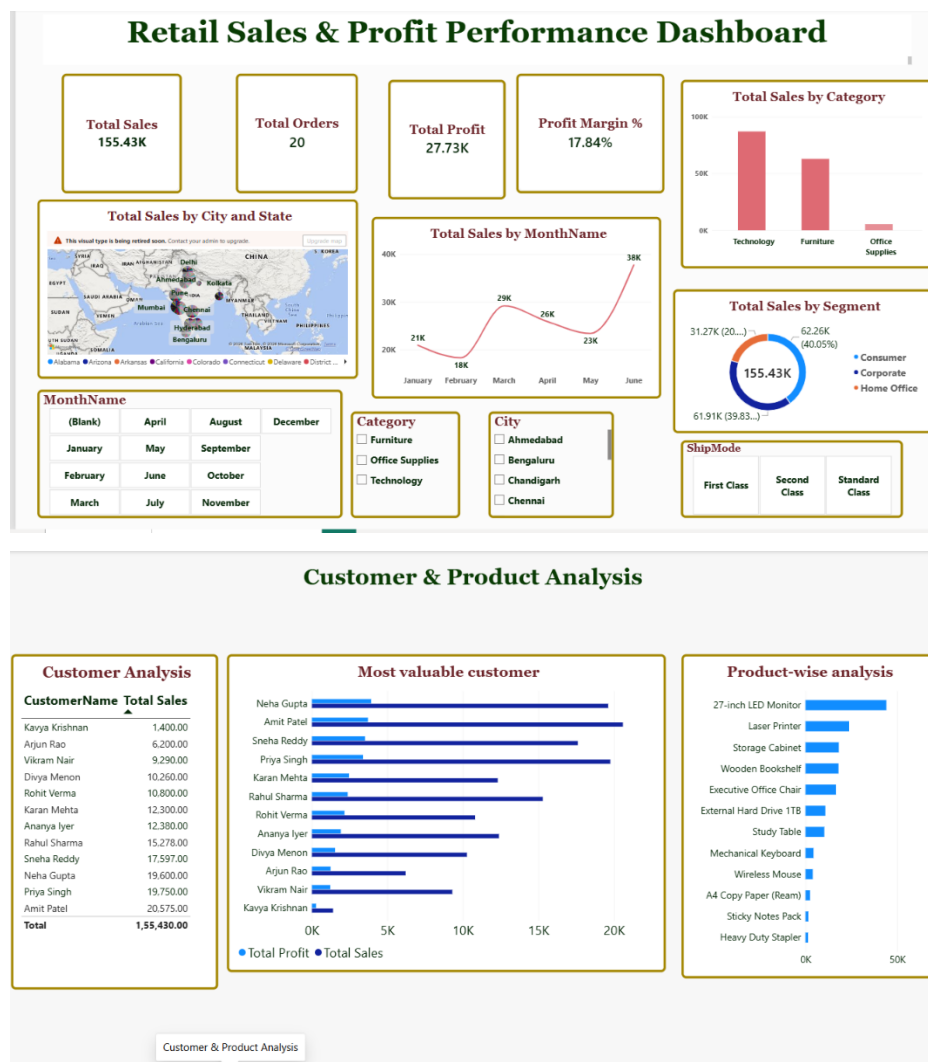
3. MySQL Database Script

- Database "retail_analytics" created with 3 related tables: Customers, Products, Orders (fact table with foreign keys).
- 12 Customers, 12 Products, and 20 Orders inserted using CREATE TABLE and INSERT statements.

```
1 • CREATE DATABASE IF NOT EXISTS retail_analytics;
2 • USE retail_analytics;
3
4 -- 1. Customers table
5 • CREATE TABLE Customers (
6     CustomerID INT AUTO_INCREMENT PRIMARY KEY,
7     CustomerName VARCHAR(100) NOT NULL,
8     Segment VARCHAR(50),
9     City VARCHAR(50),
10    State VARCHAR(50),
11    Country VARCHAR(50),
12    PostalCode VARCHAR(20)
13 );
14
15 -- 2. Products table
16 • CREATE TABLE Products (
17     ProductID INT AUTO_INCREMENT PRIMARY KEY,
18     ProductName VARCHAR(150) NOT NULL,
19     Category VARCHAR(50),
20     SubCategory VARCHAR(50),
21     UnitPrice DECIMAL(10,2)
22 );
23
24 -- 3. Orders table (fact table)
25 • CREATE TABLE Orders (
26     OrderID INT AUTO_INCREMENT PRIMARY KEY,
27     OrderDate DATE,
28     ShipDate DATE,
29     ShipMode VARCHAR(50),
30     CustomerID INT,
31     ProductID INT,
32     Quantity INT,
33     Discount DECIMAL(4,2),
34     Sales DECIMAL(10,2),
35     Profit DECIMAL(10,2),
36     CONSTRAINT fk_orders_customer FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),
37     CONSTRAINT fk_orders_product FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
38 );
```

4. Output: Retail Sales & Profit Performance Dashboard

- Total Sales Performance — KPI cards for Sales, Profit, Orders, and Profit Margin %.
- Customer Analysis — segment-wise donut chart plus customers ranking.
- Product-wise Sales Analysis — category/sub-category chart plus products ranking.
- State and City Sales Performance — filled map of sales by state/city.
- Monthly Sales Trends & interactive filters/slicers across the whole page.
- Overall Performance: Provides a quick overview of sales, orders, profit, and profit margin.
- Sales Analysis: Compares sales across categories, cities, states, customer segments, and months.
- Profit Analysis: Tracks monthly profit and identifies profitable periods.
- Customer Analysis: Identifies high-value customers based on their sales contribution.
- Product Analysis: Highlights top- and low-performing products.
- Interactive Filters: Allows users to filter data by month, category, city, and shipping mode.
- Business Insight: Helps management identify trends, evaluate product performance, and make data-driven business decisions



5. Insights & Findings

- Total Sales Rs.1,55,430 | Total Profit Rs.27,726.15 | Profit Margin 17.84% | 20 Orders | Avg Order Value Rs.7,771.50.
- Technology is the top category (56% of sales), led by the 27-inch LED Monitor and Laser Printer.
- Consumer and Corporate segments are almost evenly split (40% each); Home Office is smallest (20%).
- Top customers: Amit Patel, Priya Singh, Neha Gupta — together ~39% of total revenue from just 3 of 12 customers.
- Sales peaked in June (Rs.37,740, driven by a large Monitor order) and dipped lowest in February (Rs.18,300).

6. Conclusion

- Successfully combined a relational MySQL database and a Kaggle flat file inside one Power BI report.
- MySQL data was connected live via Power BI's native connector; the CSV was imported via Text/CSV.
- Data cleaning, relationship modeling, DAX measures, and an interactive dashboard were built on the combined model.
- The result gives management a centralized, decision-ready view of sales performance.

7. Learning Outcomes

- Designing and populating a relational MySQL database using SQL.
- Connecting Power BI to MySQL using the native connector and a database driver.
- Importing and cleaning a flat file (CSV) from an external source like Kaggle.
- Building relationships and DAX measures in the Power BI data model.
- Designing an interactive, insight-driven dashboard with slicers and visuals.